No Positive Margin from Pencils: Xiao's Lemma, a Rank-r Margin Identity, and the Nef Barrier in Parshin–Moret-Bailly Height Arguments

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Abstract

Parshin–Moret-Bailly height arguments seek an upper bound for the arithmetic self-intersection $\overline{\omega}_P^2$ of the relative dualizing sheaf on the fibers of a fibration $\phi\colon X \to Y$ over a curve. A recent implementation of this strategy (Soulé, arXiv:2611.05217, version 1) obtains its key estimate from a pencil: two sections of an auxiliary line bundle L without common zero, Minkowski's theorem on successive minima, and a transfer from classical to arithmetic degrees. The leading coefficient of the resulting inequality is the geometric margin $M = (-L^2 + m \deg \phi_* L)/\omega^2$, where m is the fiber degree of L, and an upper bound for $\overline{\omega}_P^2$ requires $M > 0$. We show that this never happens. In rank two, the Koszul resolution of the evaluation map gives the exact identity $-L^2 + m \deg \phi_* L = -\ell(Z) - 2m \deg V \leq 0$ (Xiao's lemma for pencils), where Z and V are the finite and vertical parts of the base locus. In rank r, we prove a margin identity in terms of the Bogomolov discriminant of the syzygy kernel together with a nef lemma in the spirit of Xiao, which gives $M_r \leq 0$ whenever $\phi_* L$ is semistable and strict or partial negativity in the Harder–Narasimhan-unstable case; the arithmetic side of the unstable case for $r \geq 3$ is left open. As an application, we show that Proposition 9 vi) of arXiv:2611.05217v1 is false: in the Kodaira fibration used there one has $k = 4$, at most two of the four square roots over each exceptional point are effective, and $\deg \phi_* L = -7k + 4k_{\mathrm{eff}} \leq -20 < -3k$. The step that fails uses the converse of the implication proved in Proposition 5 ii) of that preprint. We add brief remarks on cross terms, on single sections and jets along curves, and on source bundles not pulled back from the base. No statement about the *abc* conjecture is made.

1 Introduction

Parshin [11] proposed an arithmetic analogue of the Bogomolov–Miyaoka–Yau inequality for complex surfaces (for the inequalities of Bogomolov and Miyaoka see [2, 8]): an upper bound for the arithmetic self-intersection of the relative dualizing sheaf of an arithmetic surface in terms of the discriminant of the base. Parshin and Moret-Bailly [11, 9] showed that such a bound, applied to the fibers of a suitable fibration, has strong Diophantine consequences, including effective Mordell-type statements and forms of the *abc* conjecture (see also [12]). We call arguments of this kind *Parshin–Moret-Bailly (PMB) height arguments*.

In a recent preprint [13] (arXiv:2611.05217, version 1, submitted on 3 November 2026), Soulé implements this strategy for a Kodaira fibration $\phi\colon X \to Y$ with fibers of genus 6 over an étale cover of a curve of genus 2 [6, 7]. The upper bound for $\overline{\omega}_P^2$ on the fiber X_P comes from a *pencil*: an auxiliary line bundle L on X, two sections of L_P without common zero [13, Prop. 3], Minkowski's theorem on successive minima, and a transfer from classical to arithmetic

degrees [13, Cor. 1]. The leading coefficient of the resulting inequality is a purely geometric number, which we call the *margin*,

$$M(L) = \frac{-L^2 + m \deg \phi_* L}{\omega^2}, \tag{1}$$

where m is the degree of L on a fiber and $\omega = \omega_{X/Y}$. The argument bounds $\overline{\omega}_P^2$ from above if and only if $M(L) > 0$ (Section 2). In [13] the positivity of $M(L)$ rests on the lower bound $\deg \phi_* L \geq -3k$ of [13, Prop. 9 vi)].[1]

This note shows that the margin of a pencil is never positive and draws the resulting consequence for [13].

Theorem A (Theorem 3.1)**.** *Let $\phi\colon X \to Y$ be a fibration of a smooth projective surface over a smooth projective curve with integral fibers, and let L be a line bundle with $h^0(L_\eta) = 2$ whose linear system on the generic fiber is base-point-free. Let $L(-V)\otimes\mathcal{I}_Z$ be the image of $\phi^*\phi_* L \to L$. Then*

$$-L^2 + m \deg \phi_* L = -\ell(Z) - 2m \deg V \leq 0.$$

Hence $M(L) \leq 0$, with equality only if the evaluation map is surjective.

This is Xiao's lemma for pencils [14] in exact form; no stability hypothesis is needed.

Theorem B (Theorem 4.1, Corollary 5.2, Proposition 5.3)**.** *For $r = \operatorname{rk} \phi_* L \geq 2$ the relevant margin is $M_r(L) = (-L^2 + \frac{2m}{r} \deg \phi_* L)/\omega^2$, and*

$$M_r(L) = -\frac{\Delta(K) + 2rm \deg V + 2(r-1)\ell(Z)}{r\,\omega^2},$$

where $\Delta(K)$ is the Bogomolov discriminant of the syzygy kernel K. If $\phi_* L$ is semistable, then $M_r(L) \leq 0$. If $\phi_* L$ is unstable, then $M_r(L) < 0$ when the maximal destabilizing subsheaf has no horizontal base divisor, and in general every coefficient built from two Harder–Narasimhan slopes as in Proposition 5.3 (b) is nonpositive.

The semistable case rests on a nef lemma in the spirit of Xiao's method: the $\mathbb{Q}$-divisors $L - B_j - V_j - \mu_j F$ attached to the Harder–Narasimhan filtration of $\phi_* L$ are nef (Lemma 5.1). The arithmetic counterpart of the unstable case for $r \geq 3$ is left open (Open problem 5.5).

Theorem C (Theorems 6.5 and 6.6)**.** *In the Kodaira fibration of [13] one has $k = 4$, at most two of the four square roots of the hyperelliptic class over each exceptional point are effective, and*

$$\deg \phi_* L = -7k + 4k_{\mathrm{eff}} \leq -20 < -12 = -3k,$$

where $4k_{\mathrm{eff}} \leq 8$ is the number of points of Y over which the relevant square root is effective. Hence [13, Prop. 9 vi)] is false, and the leading coefficient of [13, (36)] is at most $-5/6$. A second route gives the same conclusion: the generic linear system is base-point-free for every choice of the component Y and of the square root $\mathcal{L}$, and Theorem A yields $\deg \phi_* L \leq -10k/3 < -3k$; base-point-freeness is the property used in [13, (28)] to apply [13, Prop. 3].

The step that fails in the proof of [13, Prop. 9] is inequality (24) of that paper. It uses the implication "P_1 exceptional $\Rightarrow h^0(\lambda) = 1$", whereas [13, Prop. 5 ii)] proves the converse (Remark 6.7). Section 7 records briefly why the same sign appears for cross terms, for single sections and jets evaluated along curves, and why a positive term can only come from a bundle that is not pulled back from the base.

[1] We cite [13] by the numbers printed in version 1. In the TEX source of version 1, the statements printed as Propositions 3, 4, 5, 8, and 9 carry the labels `prop12`, `prop3`, `prop13`, `prop5`, and `prop6`, while Lemma 3 carries the label `lm15`. Equation numbers such as (24) and (28)–(36) are set by hand in the source and agree with the printed ones.

Claim Boundary. This note makes no statement about the *abc* conjecture, about Parshin's conjecture, or about the truth of Theorems 1, A or B of [13] as statements. It shows that the pencil step of a PMB argument cannot produce a positive margin under the hypotheses stated below, and that the proof of [13, Theorem 1] given in version 1 is incomplete at Proposition 9 vi). All references to [13] concern version 1; a later version may differ. The intersection numbers of [13, Prop. 8 and Prop. 9 ii), iii)] that we use agree with our own computation.

2 Setting and the Margin of a Pencil

2.1 Conventions

In Sections 2–5 we work over an algebraically closed field of characteristic 0. Let $\phi\colon X \to Y$ be a surjective morphism from a smooth projective surface onto a smooth projective curve whose fibers are all integral. Let F be the class of a fiber, η the generic point of Y, and $\omega = \omega_{X/Y}$, and assume $\omega^2 > 0$. Smooth fibrations, in particular Kodaira fibrations, satisfy these hypotheses. A divisor is *vertical* if its components lie in fibers and *horizontal* if no component does. Because the fibers are integral, every effective vertical divisor is of the form $V = \sum_y n_y X_y$ with $n_y \geq 0$; we set $\deg V = \sum_y n_y$, so that $V \equiv (\deg V)F$, $V^2 = 0$, and $V \cdot F = 0$.

For a line bundle L on X, set $m = L \cdot F \geq 1$ and $E = \phi_* L$. As a torsion-free sheaf on the smooth curve Y, E is locally free of rank $r = h^0(X_\eta, L_\eta)$; we write $\mu(E) = \deg E/r$. Note that $L \cdot V = m \deg V$. We will often assume:

(BPF) The linear system $|L_\eta|$ on the generic fiber has no base point.

Lemma 2.1. *Let $0 \neq E' \subseteq E$ be a subsheaf. The image of the composite $\phi^* E' \to \phi^* E \to L$ is $L(-B' - V') \otimes \mathcal{I}_{Z'}$ with $B' \geq 0$ horizontal, $V' \geq 0$ vertical, and Z' a subscheme of finite length. The restriction of $L(-B')$ to X_η has a nonzero section. Under* (BPF) *one has $B' = 0$ for $E' = E$.*

Proof. On a smooth surface, every nonzero subsheaf of a line bundle L is of the form $L(-D) \otimes \mathcal{I}_{Z'}$ with $D \geq 0$ and Z' of finite length; decompose $D = B' + V'$ into its horizontal and vertical parts. Over η the map is the evaluation of the sections in $E'_\eta \subseteq E_\eta = H^0(X_\eta, L_\eta)$, so its image is the subsheaf generated by these sections; its divisorial base locus is $B'|_{X_\eta}$, and any nonzero section in E'_η defines a section of $L(-B')|_{X_\eta}$. For $E' = E$ this base locus is empty under (BPF). □

Definition 2.2. Set $r_1 = -L^2/\omega^2$ and $q = -\deg E/\omega^2$. The *margin* of L is

$$M_r(L) = r_1 - \frac{2m}{r}\, q = \frac{-L^2 + \frac{2m}{r} \deg E}{\omega^2}.$$

For $r = 2$ we write $M(L) = M_2(L) = r_1 - mq$, in accordance with (1).

Lemma 2.3. *One has $M_r(L) = -(L - \mu(E)F)^2/\omega^2$ and $M_r(L + dF) = M_r(L)$ for every integer d.*

Proof. We have $(L - \mu F)^2 = L^2 - 2m\mu = L^2 - \frac{2m}{r} \deg E$. Twisting by dF changes L^2 by $2md$ and, by the projection formula, $\deg E$ by rd. □

2.2 The Pencil Chain

Now let $\phi\colon X \to Y$ be defined over a number field K with geometrically integral fibers; the conventions above apply to $X_{\overline{K}} \to Y_{\overline{K}}$, which leaves intersection numbers and degrees unchanged. For a finite extension F/K of degree d and a point $P \in Y(F)$, let X_P be the fiber, equip ω and L with adelic metrics as in [13, §1.3], and let $H_P = H^0(X_P, L_P)$ carry the sup-norms. For $r = 2$, the proof of [13, Theorem 1] in [13, §6.3] combines three steps:

(i) *Pencil inequality* [13, Prop. 3, (28), (29)]: if $s_1, s_2 \in H_P$ have no common zero, then $-\overline{L}_P^2 \le m\,(h(s_1)+h(s_2)) + O(d)$.

(ii) *Minkowski* [13, (2), (30)]: there are linearly independent $s_1, s_2 \in H_P$ with $h(s_1)+h(s_2) \le -\widehat{\deg}(H_P) + \delta(F) + O(d)$.

(iii) *Transfer* [13, (32), Cor. 1, (33), (35)]: for P outside a finite set and every $\varepsilon > 0$, $\widehat{\deg}(H_P) = -q\,\overline{\omega}_P^2 + O(\varepsilon\,\overline{\omega}_P^2) + O_\varepsilon(d)$ and $\overline{L}_P^2 = -r_1\,\overline{\omega}_P^2 + O(\varepsilon\,\overline{\omega}_P^2) + O_\varepsilon(d)$.

Together they give

$$(M(L) - O(\varepsilon))\,\overline{\omega}_P^2 \le m\,\delta(F) + O_\varepsilon(d). \tag{2}$$

For general r, two sections realizing the first two successive minima of H_P satisfy $\lambda_1 + \lambda_2 \le \frac{2}{r}(\lambda_1 + \cdots + \lambda_r)$, and Minkowski's second theorem bounds the sum by $-\widehat{\deg}(H_P)$ up to a multiple of $\delta(F)$ and $O(d)$. If these two sections have no common zero, the same steps yield (2) with $M_r(L)$ in place of $M(L)$ and a modified constant in front of $\delta(F)$.

Remark 2.4. Inequality (2) bounds $\overline{\omega}_P^2$ from above if and only if $M_r(L) > 0$; for $M_r(L) \le 0$ it is vacuous. In [13] one has $r = 2$, $m = 6$, $L^2 = -20k$, and $\omega^2 = 12k$, so $r_1 = 5/3$; Proposition 9 vi) gives $q \le 1/4$, hence $M(L) \ge 1/6$ and the constant $36 = 6/(1/6)$ of [13, Theorem 1]. The ε- and $O(d)$-terms do not affect the sign. All results below are statements about the classical invariants L^2, $\deg \phi_* L$, and ω^2; in this sense they are independent of the arithmetic details of the chain.

3 Rank Two: Xiao's Lemma for Pencils

Theorem 3.1. *In the setting of Section 2, assume $r = 2$ and* (BPF)*, and let $L(-V) \otimes \mathcal{I}_Z$ be the image of the evaluation map $\phi^* E \to L$ (Lemma 2.1). Then*

$$-L^2 + m \deg E = -\ell(Z) - 2m \deg V \le 0. \tag{3}$$

In particular, $\deg \phi_* L \le L^2/m$*, with equality if and only if $\phi^*\phi_* L \to L$ is surjective.*

Proof. Set $A = L(-V)$, so that the evaluation map induces a surjection $\phi^* E \to A \otimes \mathcal{I}_Z$, and let K be its kernel.

Step 1: K is a line bundle. At every point, the ideal $\mathcal{I}_Z$ is a torsion-free module over a two-dimensional regular local ring, hence of depth at least one and, by the Auslander–Buchsbaum formula, of projective dimension at most one. The kernel of a surjection from a locally free sheaf onto a sheaf of projective dimension at most one is locally free; thus K has rank one.

Step 2: Chern classes of the outer terms. Since $\phi^* E$ is pulled back from the curve Y, we have $c_1(\phi^* E) \equiv (\deg E) F$ and $c_2(\phi^* E) = \phi^* c_2(E) = 0$. From $0 \to \mathcal{I}_Z \to \mathcal{O}_X \to \mathcal{O}_Z \to 0$ and $\mathrm{ch}(\mathcal{O}_Z) = \ell(Z)[\mathrm{pt}]$ one obtains $c_1(\mathcal{I}_Z) = 0$ and $c_2(\mathcal{I}_Z) = \ell(Z)$; tensoring the rank-one sheaf $\mathcal{I}_Z$ by A gives $c_1(A \otimes \mathcal{I}_Z) = A$ and $c_2(A \otimes \mathcal{I}_Z) = \ell(Z)$.

Step 3: Whitney formula. We have $c_1(K) \equiv (\deg E) F - A$, and

$$0 = c_2(\phi^* E) = c_1(K) \cdot A + \ell(Z) = m \deg E - A^2 + \ell(Z),$$

using $F \cdot A = m$ (since $F \cdot V = 0$).

Step 4. We have $A^2 = (L - V)^2 = L^2 - 2m \deg V$, since $V^2 = 0$ and $L \cdot V = m \deg V$. Substituting gives (3). Equality $\deg E = L^2/m$ holds if and only if $Z = \emptyset$ and $V = 0$, that is, if and only if the evaluation map is surjective. □

Corollary 3.2. *Under the hypotheses of Theorem 3.1,*

$$M(L) = -\frac{\ell(Z) + 2m \deg V}{\omega^2} = -\frac{(L - \mu(E) F)^2}{\omega^2} \le 0.$$

The pencil chain (2) *is therefore vacuous for every line bundle of rank two with base-point-free generic linear system on every fibration with integral fibers.*

Remark 3.3. (a) Theorem 3.1 requires no stability hypothesis on ϕ_*L; it is the exact rank-two form of Xiao's method [14]. Equivalently, $(L-\mu(E)F)^2 = \ell(Z) + 2m\deg V \geq 0$.

(b) For $r=2$, no cleverer choice of the two sections helps. Minkowski's second theorem determines $\lambda_1+\lambda_2$ up to terms of order $\delta(F)+d$, so together with step (iii) every pair of linearly independent sections satisfies $h(s_1)+h(s_2) \geq q\,\overline{\omega}_P^2 - O(\varepsilon\,\overline{\omega}_P^2 + \delta(F) + d)$. The best coefficient the chain can achieve is $r_1 - mq = M(L) \leq 0$.

(c) Twisting by vertical divisors does not help either: $M(L+dF) = M(L)$ by Lemma 2.3.

4 Rank r: A Margin Identity

For a locally free sheaf G of rank n on X, let $\Delta(G) = 2n\,c_2(G) - (n-1)\,c_1(G)^2 = c_1(G)^2 - 2n\,\mathrm{ch}_2(G)$ denote its Bogomolov discriminant; $\Delta(G)=0$ for line bundles.

Theorem 4.1. *In the setting of Section 2, assume* (BPF) *and $r \geq 2$. Let $L(-V)\otimes\mathcal{I}_Z$ be the image of $\phi^*E \to L$ and K its kernel. Then K is locally free of rank $r-1$, and*

$$\Delta(K) = r\,L^2 - 2m\deg E - 2rm\deg V - 2(r-1)\ell(Z), \tag{4}$$

or equivalently

$$M_r(L) = -\frac{\Delta(K) + 2rm\deg V + 2(r-1)\ell(Z)}{r\,\omega^2}.$$

Proof. Local freeness follows as in Step 1 of Theorem 3.1. Set $A = L(-V)$. From $c(\phi^*E) = c(K)\,c(A\otimes\mathcal{I}_Z)$ and Step 2 above, we have $c_1(K) \equiv (\deg E)F - A$ and $0 = c_2(K) + c_1(K)\cdot A + \ell(Z)$, whence

$$c_1(K)^2 = A^2 - 2m\deg E, \qquad c_2(K) = A^2 - m\deg E - \ell(Z).$$

Therefore $\Delta(K) = 2(r-1)c_2(K) - (r-2)c_1(K)^2 = r\,A^2 - 2m\deg E - 2(r-1)\ell(Z)$, and substituting $A^2 = L^2 - 2m\deg V$ gives (4). Dividing by $-r\,\omega^2$ yields the expression for $M_r(L)$. For $r=2$ the kernel is a line bundle, $\Delta(K)=0$, and (4) reduces to (3). □

Corollary 4.2. *If, in addition, K is μ-semistable with respect to some ample class on X, then Bogomolov's inequality $\Delta(K) \geq 0$ [2] (see also [5, §3.4]) gives $M_r(L) \leq 0$.*

Remark 4.3. Semistability of the syzygy kernel is not automatic, and for $r \geq 3$ the estimate $L^2 \geq \frac{2m}{r}\deg\phi_*L$ is a slope inequality that is derived in the literature under stability hypotheses (compare [14, 3]). Section 5 replaces the hypothesis on K by semistability of ϕ_*L on the base curve and also treats the unstable case.

5 The Nef Lemma and the Harder–Narasimhan Dichotomy

Let $0 = E_0 \subsetneq E_1 \subsetneq \cdots \subsetneq E_n = E$ be the Harder–Narasimhan filtration of $E = \phi_*L$ [4], with slopes $\mu_j = \mu(E_j/E_{j-1})$, so that $\mu_1 > \cdots > \mu_n$ and μ_j is the minimal slope of E_j. By Lemma 2.1, the image of $\phi^*E_j \to L$ is $L(-B_j - V_j)\otimes\mathcal{I}_{Z_j}$; if (BPF) holds, then $B_n = 0$, $V_n = V$, and $Z_n = Z$. Set

$$N_j = L - B_j - V_j - \mu_j F \in \mathrm{Pic}(X)\otimes\mathbb{Q}.$$

Lemma 5.1 (nef lemma)**.** *For every j, the $\mathbb{Q}$-divisor N_j is nef. Hypothesis* (BPF) *is not needed.*

Proof. Let $C \subset X$ be an irreducible curve. If C is vertical, then C is a fiber and $N_j \cdot C = (L - B_j) \cdot F \geq 0$, because $L(-B_j)|_{X_\eta}$ has a nonzero section (Lemma 2.1). Now let C be horizontal, $\nu\colon \widetilde{C} \to C$ its normalization, and $f = \phi \circ \nu\colon \widetilde{C} \to Y$, which is a finite morphism of degree $d_C = C \cdot F \geq 1$. The map $\phi^* E_j \to L(-B_j - V_j) \otimes \mathcal{I}_{Z_j}$ is surjective onto $L(-B_j - V_j)$ outside the finite set Z_j, in particular at the generic point of C. Pulling back to $\widetilde{C}$ gives a nonzero homomorphism $f^* E_j \to \nu^* L(-B_j - V_j)$ whose image Q is an invertible quotient of $f^* E_j$ contained in $\nu^* L(-B_j - V_j)$. Hence

$$(L - B_j - V_j) \cdot C = \deg \nu^* L(-B_j - V_j) \geq \deg Q \geq \mu_{\min}(f^* E_j) = d_C\, \mu_j.$$

The last equality holds because in characteristic 0 the pullback of the Harder–Narasimhan filtration under a finite morphism of smooth curves coincides with the Harder–Narasimhan filtration of the pullback, with slopes multiplied by the degree (see [5, §3.2]). Since $F \cdot C = d_C$, this gives $N_j \cdot C \geq 0$. □

Corollary 5.2 (semistable case)**.** *Assume* (BPF) *and that $E = \phi_* L$ is semistable. Then $N = L - V - \mu(E)F$ is nef, and*

$$M_r(L) = -\frac{N^2 + 2m \deg V}{\omega^2} \leq 0.$$

No hypothesis on the stability of the fibers, of their linear systems, or of the syzygy kernel is needed.

Proof. Here $n = 1$, $E_1 = E$, $B_1 = 0$, $V_1 = V$, and $\mu_1 = \mu(E)$, so $N = N_1$ is nef by Lemma 5.1 and hence $N^2 \geq 0$. Since $L - \mu(E)F = N + V$, $N \cdot V = m \deg V$, and $V^2 = 0$, we obtain $(L - \mu(E)F)^2 = N^2 + 2m \deg V$; the claim follows from Lemma 2.3. □

Proposition 5.3 (unstable case)**.** *Assume* (BPF) *and that $E = \phi_* L$ is unstable, $n \geq 2$.*

(a) If $B_1 = 0$, then $M_r(L) \leq -2m(\mu_1 - \mu(E) + \deg V_1)/\omega^2 < 0$.

(b) For all indices a, b with $B_b = 0$,

$$L^2 \geq m(\mu_a + \mu_b) + m \deg V_b, \qquad \text{that is} \qquad \frac{-L^2 + m(\mu_a + \mu_b)}{\omega^2} \leq -\frac{m \deg V_b}{\omega^2} \leq 0.$$

Such an index b always exists, since $B_n = 0$.

Proof. (a) We have $L - \mu(E)F = N_1 + V_1 + (\mu_1 - \mu(E))F$. Expanding with $N_1 \cdot V_1 = m \deg V_1$, $N_1 \cdot F = m$ (as $B_1 = 0$), and $V_1^2 = V_1 \cdot F = F^2 = 0$ gives $(L - \mu(E)F)^2 = N_1^2 + 2m \deg V_1 + 2m(\mu_1 - \mu(E))$. Since $N_1^2 \geq 0$ and $\mu_1 > \mu(E)$ for unstable E, the claim follows from Lemma 2.3.

(b) The divisors N_b and N_a are nef and B_a, V_a are effective, so $N_b \cdot (N_a + B_a + V_a) \geq 0$. Here $N_a + B_a + V_a = L - \mu_a F$ and $N_b = L - V_b - \mu_b F$, and the intersection product equals $L^2 - m\mu_a - m\mu_b - m \deg V_b$. □

Remark 5.4. In case (a), the margin that the chain of Section 2.2 produces with Minkowski's theorem on the full lattice is strictly negative. The coefficients in (b) are those that the chain would produce if the heights of the two sections were governed by the slopes μ_a and μ_b; for instance, Corollary 1 of [13], applied to the sublattice coming from E_1, gives a section s_1 with $h(s_1) \leq (-\mu_1/\omega^2 + \varepsilon)\,\overline{\omega}_P^2 + c\,\delta(F) + O_\varepsilon(d)$ for a constant c. Thus every coefficient that the chain can obtain from geometric subbundles in this way is nonpositive. The condition $B_b = 0$ in (b) reflects the pencil step: every section coming from E_j vanishes along $B_j \cap X_P$, so two sections taken from the same step E_a with $B_a \neq 0$ always have a common zero on X_P. If B_a and B_b merely have no common component, their restrictions to X_P are disjoint for all but finitely many P, and the same nonpositivity follows from Proposition 7.1 with $L_1 = L_2 = L$.

Open problem 5.5 (arithmetic anomaly, $r \geq 3$)**.** Let $r \geq 3$ and assume that $\phi_* L$ is unstable. We do not know whether, for points P of large height, an upper bound for $h(s_1)+h(s_2)$ over pairs of sections without common zero can be obtained that lies below what the geometric slopes μ_a predict. Excluding this would require a comparison between the arithmetic Harder–Narasimhan polygon of the adelic lattice H_P and the geometric Harder–Narasimhan polygon of $\phi_* L$, scaled by $\overline{\omega}_P^2/\omega^2$. We leave this question open. For $r = 2$ it does not arise (Remark 3.3 (b)), and for semistable $\phi_* L$ it is answered by Corollary 5.2.

6 Application to Soulé, arXiv:2611.05217v1

6.1 The Construction

We recall the Kodaira fibration of [13, §§3, 5], which follows [6, 7]. Let Y_0 be a curve of genus 2 and $\chi\colon Y_1 \to Y_0$ a nontrivial étale double cover with covering involution ν; Y_1 has genus 3 and is hyperelliptic [13, Prop. 4 i)]. Let ι be its hyperelliptic involution and H the hyperelliptic class; then $\omega_{Y_1} \cong \mathcal{O}(2H) \cong \mathcal{O}(4W)$ for a Weierstrass point W. Restricting the relation $\mathcal{O}(\Gamma - 2\Delta_1) = (1 \times \psi_1)^*\mathcal{P} + \pi_2^* L$ of [13, §5.1] to $Y_1 \times \{x\}$ shows that $\psi_1\colon Y_1 \to J_1$ is given by $x \mapsto [\nu x - x]$. The curve Y is a connected component of the fiber product Z of ψ_1 with multiplication by 2 on J_1, and $\psi\colon Y \to Y_1$ is finite étale of degree k.

On $S = Y_1 \times Y$ with projections pr_1 and $p\colon S \to Y$, let D_1 and D_2 be the graphs of ψ and $\nu \circ \psi$; they are disjoint because ν has no fixed point. Let $\mathcal{L} \in \mathrm{Pic}(S)$ with $2\mathcal{L} = \mathcal{O}_S(D_1 + D_2)$ [7], let $\rho\colon X \to S$ be the double cover branched along $D_1 + D_2$, and set $\phi = p \circ \rho$; this morphism is smooth with fibers of genus 6. Let $E_i = \rho^{-1}(D_i)_{\mathrm{red}}$, so that $\rho^* D_i = 2E_i$, and set $B_i = \mathcal{O}_X(E_i)$. For $y \in Y$ set $P_1 = \psi(y)$, $P_2 = \nu P_1$, and $\lambda_y = \mathcal{L}|_{Y_1 \times \{y\}}$, a square root of $\mathcal{O}(P_1 + P_2)$. The point P_1 is *exceptional* if $h^0(P_1 + P_2) = 2$; equivalently, $P_2 = \iota P_1$ [13, Prop. 5 iii)]. There are four exceptional points, and $\Psi = \psi^{-1}(\text{exceptional points})$ has $4k$ elements.

Let $G = \phi_1^* \mathcal{O}(W)$ with $\phi_1 = \mathrm{pr}_1 \circ \rho$. The lattice spanned by G, B_1, B_2, F has the intersection numbers in Table 1; they are those of [13, Prop. 8] together with $B_i \cdot F = 1$ (E_i is a section) and $G \cdot F = 2$ (ϕ_1 has degree two on fibers). The Riemann–Hurwitz formula gives $\omega = 4G + B_1 + B_2$ and $\omega^2 = 12k$. Soulé's auxiliary bundle is

$$L = \omega - 2B_1 - 2B_2 = 4G - B_1 - B_2, \qquad m = 6, \qquad L^2 = -20k,$$

and $h^0(L_y) = 2$ for general y [13, Prop. 9 ii), iii), v)].

	G	B_1	B_2	F
G	0	k	k	2
B_1	k	$-2k$	0	1
B_2	k	0	$-2k$	1
F	2	1	1	0

Table 1: Intersection numbers on the Kodaira fibration of [13].

6.2 The Count

Lemma 6.1 ($k = 4$)**.** *Let $G_2 \subseteq J_1[2]$ be the monodromy group of the $J_1[2]$-torsor $Z \to Y_1$. Then $G_2 = (\nu_* - 1)\, J_1[2]$ and $|G_2| = 4$. Consequently, $k = 4$ for every choice of Y_0, of χ, and of the component Y.*

Proof. The torsor is the pullback of $[2]\colon J_1 \to J_1$ along ψ_1, so its monodromy is $\pi_1(Y_1) \to H_1(Y_1, \mathbb{Z}) \xrightarrow{(\psi_1)_*} H_1(J_1, \mathbb{Z}) \to J_1[2]$ with $(\psi_1)_* = \nu_* - 1$, and every connected component has

degree $|G_2|$ over Y_1. The order can be computed over $\mathbb{C}$. Since the mapping class group acts transitively on the nonzero classes in $H^1(Y_0, \mathbb{Z}/2)$, we may take $\pi_1(Y_0) = \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2]\rangle$ and assume that χ is given by the character with $a_1 \mapsto 1$ and $b_1, a_2, b_2 \mapsto 0$. With the Schreier transversal $\{1, a_1\}$, the subgroup $\pi_1(Y_1)$ is generated by $s = a_1^2$, b_1, a_2, b_2 and their conjugates b_1', a_2', b_2' by a_1. The defining relator and its conjugate both map to $b_1' = b_1$ in the abelianization, so $H_1(Y_1, \mathbb{Z})$ is free on $s, b_1, a_2, b_2, a_2', b_2'$. The involution ν corresponds to conjugation by a_1; on homology it fixes s and b_1 and exchanges $a_2 \leftrightarrow a_2'$ and $b_2 \leftrightarrow b_2'$. The image of $\nu_* - 1$ is spanned by $a_2' - a_2$ and $b_2' - b_2$, which extend to a basis, so its reduction modulo 2 has order 4. □

Lemma 6.2 (effective square roots)**.** *Let P_1 be exceptional. The classes λ_y, $y \in \psi^{-1}(P_1)$, form a coset of G_2 in the set of square roots of H. Every coset of G_2 in this set contains at most two effective classes.*

Proof. For $(x, z) \in Z$, the class $x + z \in \mathrm{Pic}^1(Y_1)$ is a square root of $x + \nu x$, since $2(x + z) = 2x + \nu x - x$. The map $y \mapsto \lambda_y - (\psi(y) + \psi_2(y))$, with $\psi_2 \colon Y \to J_1$ denoting the second projection, takes values in $J_1[2]$ and is therefore constant on the connected curve Y. Over P_1 the points of Y form a G_2-orbit in the fiber of Z, so the classes λ_y form a coset of G_2. An effective square root of H has the form $\mathcal{O}(Q)$ with $2Q \in |H|$, and thus corresponds to a Weierstrass point. Let $W \neq W'$ be effective in the same coset. Then $W' - W = (\nu - 1)g$ for some $g \in J_1[2]$, hence $(1 + \nu)(W' - W) = (\nu^2 - 1)g = 0$ and $W' + \nu W' \sim W + \nu W$. If these divisors were distinct, $W + \nu W$ would move in a pencil and hence lie in $|H|$, giving $\nu W = \iota W = W$, which is impossible because ν is fixed-point-free. Hence $W' = \nu W$, and a coset contains at most the two effective roots $W, \nu W$. □

Lemma 6.3 (torsion of length one)**.** *The torsion of $R^1 p_* \mathcal{L}$ is supported precisely at the points y with $h^0(\lambda_y) \neq 0$, and has length one at each of them.*

Proof. Generically $h^0(\lambda_y) = 0$ [13, Prop. 5 ii)], so $p_* \mathcal{L} = 0$. Let π be a uniformizer at y. The exact sequence $0 \to \mathcal{L} \xrightarrow{\pi} \mathcal{L} \to \lambda_y \to 0$ gives $R^1 p_* \mathcal{L}[\pi] \cong H^0(\lambda_y)$, so the torsion T_y is nonzero precisely when $h^0(\lambda_y) \neq 0$, and it is cyclic because $h^0(\lambda_y) \leq 1$. Likewise, $R^1 p_* \mathcal{L}[\pi^2]$ is isomorphic to the space of sections of $\mathcal{L}$ over $Y_1 \times \operatorname{Spec} \mathcal{O}_{Y,y}/\pi^2$, so $\ell(T_y) \geq 2$ if and only if the section s_W of $\lambda_y = \mathcal{O}(W)$ extends to first order. This holds if and only if $v \cup s_W = 0$ in $H^1(\mathcal{O}(W))$, where $v \in H^1(\mathcal{O}_{Y_1})$ is the tangent vector of $y \mapsto \lambda_y$; that is, if and only if v lies on the line $d\mathrm{AJ}_W(T_W Y_1)$. Under $H^1(\mathcal{O}_{Y_1}) \cong H^0(\omega_{Y_1})^\vee$, this line is spanned by the evaluation at W. Since $2\lambda_y = \mathcal{O}(\psi(y) + \nu\psi(y))$ and ψ is étale, $2v$ is the functional $\alpha \mapsto \alpha_{P_1}(t) + \alpha_{P_2}(d\nu\, t)$ for a nonzero tangent vector t at P_1. At an exceptional point $P_2 = \iota P_1$, since ι commutes with ν, the involution $\nu\iota$ fixes P_2 and has derivative -1 there, so $d\nu = -d\iota$ at P_1; as $\iota^* \alpha = -\alpha$ for all $\alpha \in H^0(\omega_{Y_1})$, we obtain $2v = 2 \operatorname{ev}_{P_1, t} \neq 0$. The evaluations at P_1 and W are proportional only if the canonical map identifies P_1 and W, that is, if $P_1 \in \{W, \iota W\} = \{W\}$. However, $P_1 = W$ would give $W \sim P_2$ and hence $P_2 = P_1$, a contradiction. Thus $\ell(T_y) = 1$. □

Proposition 6.4 (degree formula)**.** *Let T be the number of points $y \in Y$ with $h^0(\lambda_y) \neq 0$; all of them lie in Ψ. Then*

$$\deg \phi_* L = -7k + T.$$

Proof. Set $K_1 = \mathrm{pr}_1^* \omega_{Y_1} = \omega_{S/Y}$. The canonical bundle formula for double covers [1, I.17] gives $\omega_{X/Y} = \rho^*(K_1 + \mathcal{L})$, and $B = 2B_1 + 2B_2 = \rho^*(D_1 + D_2) = \rho^*(2\mathcal{L})$. Hence $L = \rho^*(K_1 - \mathcal{L})$, and by $\rho_* \mathcal{O}_X = \mathcal{O}_S \oplus \mathcal{L}^{-1}$ [13, (13)] we have

$$\phi_* L = p_*(K_1 - \mathcal{L}) \oplus p_*(K_1 - 2\mathcal{L}).$$

Second summand. Since $x + \iota x \in |H|$ for every x, the seesaw principle gives $\mathcal{O}_S(D_i + \iota D_i) \cong \mathrm{pr}_1^* \mathcal{O}(H) \otimes p^* M_i$ with $M_i \in \mathrm{Pic}(Y)$, where $\iota D_i = (\iota \times 1)(D_i)$. Intersecting with $\{x\} \times Y$,

which meets each of D_i and ιD_i transversally in k points and on which $\mathrm{pr}_1^*\mathcal{O}(H)$ is trivial, gives $\deg M_i = 2k$. As $K_1 = \mathrm{pr}_1^*\mathcal{O}(2H)$, we have

$$K_1 - 2\mathcal{L} = K_1 - D_1 - D_2 \cong \mathcal{O}_S(\iota D_1 + \iota D_2) \otimes p^*(-M_1 - M_2).$$

The automorphism $\iota \times 1$ of S over Y and [13, (20)] give $p_*\mathcal{O}_S(\iota D_1 + \iota D_2) \cong p_*\mathcal{O}_S(D_1 + D_2) \cong \mathcal{O}_Y$, hence $\deg p_*(K_1 - 2\mathcal{L}) = -4k$.

First summand. Relative duality gives $p_*(K_1 - \mathcal{L}) \cong \mathcal{H}om(R^1p_*\mathcal{L}, \mathcal{O}_Y)$, so $\deg p_*(K_1 - \mathcal{L}) = -(\deg R^1p_*\mathcal{L} - T)$ by Lemma 6.3. Since $p_*\mathcal{L} = 0$, Grothendieck–Riemann–Roch for p gives $\deg R^1p_*\mathcal{L} = -(\frac{1}{2}\mathcal{L}^2 - \frac{1}{2}K_1 \cdot \mathcal{L})$, and $4\mathcal{L}^2 = (D_1 + D_2)^2 = -8k$ together with $2K_1 \cdot \mathcal{L} = K_1 \cdot (D_1 + D_2) = 8k$ yields $\deg R^1p_*\mathcal{L} = 3k$. Hence $\deg p_*(K_1 - \mathcal{L}) = T - 3k$.

Adding the two degrees gives $\deg \phi_* L = T - 3k - 4k$. By [13, Prop. 5 ii)], $h^0(\lambda_y) \neq 0$ forces P_1 to be exceptional, so the points counted by T lie in Ψ. □

Theorem 6.5. *In the Kodaira fibration of [13], with $L = \omega - 2B_1 - 2B_2$,*

$$\deg \phi_* L = -7k + 4k_{\mathrm{eff}}, \qquad k = 4, \qquad 0 \leq k_{\mathrm{eff}} \leq 2,$$

where $4k_{\mathrm{eff}} = T$. In particular, $\deg \phi_* L \leq -20 < -12 = -3k$, so Proposition 9 vi) of arXiv:2611.05217v1 is false. Moreover, $q = -\deg \phi_* L/\omega^2 \geq 5/12$, and the leading coefficient $5/3 - 6q$ of [13, (36)] satisfies $5/3 - 6q \leq -5/6$.

Proof. Proposition 6.4 gives $\deg \phi_* L = -7k + T$. Lemma 6.1 gives $k = 4$, and Lemma 6.2 shows that each of the four fibers of ψ over an exceptional point contributes at most two points to T, so $T \leq 8$. Hence $\deg \phi_* L \leq -28 + 8 = -20$ and $q \geq 20/48 = 5/12$. □

Theorem 6.6 (structural form)**.** *In the setting of [13], the linear system $|L_\eta|$ is base-point-free for every choice of the component Y and of $\mathcal{L}$. Consequently, $\deg \phi_* L \leq L^2/6 = -10k/3 < -3k$ and $M(L) \leq 0$; in particular, no choice of the component Y or of $\mathcal{L}$ satisfies Proposition 9 vi).*

Proof. Step 1: fibers outside Ψ. Let $y \notin \Psi$, so that $P_2 \neq \iota P_1$. Then $h^0(\lambda_y) = 0$ by [13, Prop. 5 ii)], and Riemann–Roch gives $h^0(\omega_{Y_1} - \lambda_y) = 1$ and $h^0(\omega_{Y_1} - 2\lambda_y) = h^0(P_1 + P_2) = 1$. As $\omega_{Y_1} \cong \mathcal{O}(2H)$ and $P_i + \iota P_i \in |H|$, the unique effective divisor in $|\omega_{Y_1} - 2\lambda_y|$ is $\iota P_1 + \iota P_2$. By the decomposition of $\phi_* L$ in the proof of Proposition 6.4, $H^0(X_y, L_y)$ is spanned by $\rho^*\sigma_y$ and $t_y\, \rho^*\tau_y$, where σ_y spans $H^0(\omega_{Y_1} - \lambda_y)$, τ_y has divisor $\iota P_1 + \iota P_2$, and t_y is the section of $\rho^*\lambda_y$ that vanishes at the two ramification points. Hence $|L_y|$ has a base point if and only if σ_y vanishes at one of P_1, P_2, ιP_1, ιP_2, and by Riemann–Roch σ_y vanishes at a point Q if and only if $h^0(\lambda_y + Q) \neq 0$.

Step 2: specialization. On the open subset $Y^\circ \subseteq Y$ where $h^0(\lambda_y) = 0$, one has $h^0(\omega_{Y_1} - \lambda_y) = 1$, so $p_*(K_1 - \mathcal{L})$ is a line bundle on Y° whose formation commutes with base change. Hence, for each of the four maps $Q = \psi, \nu\psi, \iota\psi, \iota\nu\psi$ from Y to Y_1, the set of $y \in Y^\circ$ with $\sigma_y(Q(y)) = 0$ is closed in Y°. None of these sets is all of Y°: indeed, let x_0 be exceptional; by Lemmas 6.1 and 6.2 there exists $y_0 \in \psi^{-1}(x_0)$ with $h^0(\lambda_0) = 0$, where $\lambda_0 = \lambda_{y_0}$. Since $\nu x_0 = \iota x_0 \neq x_0$, the four points $Q(y_0)$ lie in $\{x_0, \iota x_0\}$, neither of them is a Weierstrass point, and $2\lambda_0 \sim x_0 + \iota x_0 \sim H$. If $\lambda_0 + q \sim a + b$ with $q \in \{x_0, \iota x_0\}$, then $2a + 2b \in |H + 2q|$. As q is not a Weierstrass point, $h^0(H + 2q) = 2$ and $|H + 2q| = |H| + 2q$, so $2a + 2b = c + \iota c + 2q$ for some point c. All multiplicities on the left-hand side are even, whereas for $c \neq \iota c$ either c or q has odd multiplicity on the right-hand side. Hence $c = \iota c$ and $a + b = c + q$, so $\lambda_0 \sim c$ is effective, a contradiction. Thus σ_{y_0} vanishes at none of the points $Q(y_0)$, the four closed subsets are finite, and by Step 1 the linear system $|L_y|$ is base-point-free for all but finitely many $y \in Y \setminus \Psi$.

Step 3: the generic fiber. For $y \in Y \setminus \Psi$ one has $h^0(L_y) = 2$, so $\phi_* L \otimes k(y) \to H^0(X_y, L_y)$ is bijective. The closure of a base point of $|L_\eta|$ is a horizontal curve along which $\phi^*\phi_* L \to L$ is not surjective; it meets every fiber X_y, and for $y \in Y \setminus \Psi$ it meets X_y in a base point of $|L_y|$,

contrary to Step 2. Hence $|L_\eta|$ is base-point-free, and Theorem 3.1 with $m = 6$ and $L^2 = -20k$ gives the claimed bound.

Step 4: relation to [13]. In [13, §6.3.3] the square root $\mathcal{L}'$ is chosen depending on the point $\bar{y}' \notin \Phi'$, and [13, Lemma 3] shows that $L'_{P'}$ has no base point. Since $h^0(L'_{P'}) = 2$ equals the rank of $\phi'_* L'$, the argument of Step 3 shows that this already forces $|L'_\eta|$ to be base-point-free; thus every bundle to which [13, (28)] is applied falls under Theorem 3.1. Steps 1–3 apply verbatim after the base change $a\colon Y' \to Y$ of [13, §6.3.1], with $\psi \circ a$ in place of ψ. Let J' be the Jacobian of Y'. Since $\mathrm{Hom}(J_1, J')$ is torsion-free, $\mathrm{Pic}(Y_1 \times Y')[2] = \mathrm{pr}_1^* J_1[2] \oplus \mathrm{pr}_2^* J'[2]$, so on fibers $\mathcal{L}'$ restricts to the classes $\lambda_{a(y')} + \varepsilon$ for a fixed $\varepsilon \in J_1[2]$. Over an exceptional point x_0, the classes $\lambda_y + \varepsilon$, $y \in \psi^{-1}(x_0)$, form a coset of G_2, so by Lemma 6.2 some $y_0 \in \psi^{-1}(x_0)$ satisfies $h^0(\lambda_{y_0} + \varepsilon) = 0$; as a is surjective, y_0 lifts to a point of $(\psi \circ a)^{-1}(x_0)$, which is all that Step 2 requires. Lemma 6.1 is used only for ψ, with $k = 4$. Base-point-freeness therefore does not depend on the pointwise choice. □

Remark 6.7 (where the proof of Proposition 9 breaks down). Inequality (24) of [13] asserts that $\deg(R^1\phi_* B)_{\mathrm{tor}} \geq 8k$. It rests on [13, Prop. 9 iv)], which states that $h^0(B_{\bar{y}}) = 3$ whenever P_1 is exceptional. Its proof uses the sentence "when P_1 is exceptional $h^0(\lambda) = 1$", with a reference to parts ii) and iii) of [13, Prop. 5]. Proposition 5 ii), however, states that $h^0(\lambda) = 0$ unless λ is a Weierstrass point, in which case $h^0(\lambda) = 1$ and $h^0(P_1 + P_2) = 2$; it thus proves "λ effective $\Rightarrow$ P_1 exceptional", the converse of the implication needed. By Lemma 6.2, the implication used fails at no fewer than $4k - 8 = 8$ of the 16 points of Ψ; there $h^0(B_{\bar{y}}) = 2$ and $h^0(L_{\bar{y}}) = 3$, so Proposition 9 iv) and v) fail at these points. Combined with [13, (16), (18), (22)], Proposition 6.4 gives $\deg(R^1\phi_* B)_{\mathrm{tor}} = 4k + T \leq 24$ instead of ≥ 32. Since a global modification of $\mathcal{L}$ by an element of $J_1[2]$ translates all cosets simultaneously, no choice of $\mathcal{L}$ alters the bound of Lemma 6.2. Theorem 1 of [13] remains open as a statement; what fails is its proof via Proposition 9 vi), and by Corollary 3.2 together with Step 4 of the proof of Theorem 6.6 the pencil step cannot be salvaged by another rank-two auxiliary bundle L' with base-point-free $|L'_\eta|$. If $|L'_\eta|$ has a base point, then $\phi_* L' \otimes k(P) \to H^0(X_P, L'_P)$ is bijective for P outside a finite set, so by the argument of Step 3 of that proof any two sections of L'_P have a common zero for all such P, and the pencil hypothesis of [13, (28)] cannot be satisfied.

7 Scope: Cross Terms, Single Sections, Source Bundles

This section records three further consequences of the nef lemma and of the identity underlying Theorem 4.1. The formal statements are geometric and proved in full; the remarks on their arithmetic meaning are interpretive.

7.1 Cross Terms

Proposition 3 of [13] is stated for two line bundles L_1, L_2 of fiber degrees d_1, d_2: $-\overline{L}_1 \cdot \overline{L}_2 \leq d_2 h(s_1) + d_1 h(s_2)$. If s_1 comes from the Harder–Narasimhan step a of $\phi_* L_1$ and s_2 from step b of $\phi_* L_2$, with heights governed by the slopes $\mu_{1,a}$ and $\mu_{2,b}$, the leading coefficient becomes

$$M(L_1, L_2) = \frac{-L_1 \cdot L_2 + d_2\mu_{1,a} + d_1\mu_{2,b}}{\omega^2} = -\frac{(L_1 - \mu_{1,a}F) \cdot (L_2 - \mu_{2,b}F)}{\omega^2}.$$

Proposition 7.1. *If the horizontal base divisors $B_{1,a}$ and $B_{2,b}$ have no common component, then $(L_1 - \mu_{1,a}F) \cdot (L_2 - \mu_{2,b}F) \geq 0$, and hence $M(L_1, L_2) \leq 0$.*

Proof. Write $L_1 - \mu_{1,a}F = N_{1,a} + B_{1,a} + V_{1,a}$ and similarly for L_2, with $N_{1,a}, N_{2,b}$ nef by Lemma 5.1. In the expansion, every term is nonnegative: the intersection of two nef divisors, of a nef and an effective divisor, $B_{1,a} \cdot B_{2,b} \geq 0$ (since they share no common component), of a horizontal and a vertical divisor, and $V_{1,a} \cdot V_{2,b} = 0$. □

7.2 Single Sections and Jets along Curves

For a single section s and an irreducible horizontal curve C not contained in the divisor of s, comparing pointwise norms with sup-norms gives a bound of the form $-\overline{L}\cdot\overline{C}\le(C\cdot F)\,h(s)$, whose geometric coefficient is $-(L-\mu_jF)\cdot C/\omega^2$ for s from E_j. Normal jets of order b along a curve C appearing in B_j with multiplicity b are sections of $(L-bC)|_C$.

Proposition 7.2. *Let C be an irreducible horizontal curve. (a) If $C\not\subseteq B_j$, then $(L-\mu_jF)\cdot C\ge 0$. (b) If C occurs in B_j with multiplicity b, then $(L-bC-\mu_jF)\cdot C\ge 0$.*

Proof. (a) We have $L-\mu_jF=N_j+B_j+V_j$ with $N_j\cdot C\ge 0$, $B_j\cdot C\ge 0$, and $V_j\cdot C\ge 0$. (b) Similarly, $L-bC-\mu_jF=N_j+(B_j-bC)+V_j$, and C is not a component of B_j-bC. □

7.3 Source Bundles Not Pulled Back from the Base

The identity of Theorem 4.1 admits a generalization to an arbitrary source bundle.

Proposition 7.3. *Let $\mathcal{F}$ be locally free of rank $r\ge 2$ on X and $\mathcal{F}\to L$ a morphism with image $A\otimes\mathcal{I}_Z$, where A is a line bundle and Z has finite length. Then the kernel K is locally free of rank $r-1$, and*

$$\Big(A-\frac{c_1(\mathcal{F})}{r}\Big)^2=\frac{\Delta(K)+2(r-1)\ell(Z)}{r}-\frac{r-1}{r^2}\,\Delta(\mathcal{F}).$$

Proof. We have $c_1(K)=c_1(\mathcal{F})-A$ and $\mathrm{ch}_2(K)=\mathrm{ch}_2(\mathcal{F})-\frac{1}{2}A^2+\ell(Z)$. Inserting these into $\Delta(K)=c_1(K)^2-2(r-1)\,\mathrm{ch}_2(K)$, using $\mathrm{ch}_2(\mathcal{F})=(c_1(\mathcal{F})^2-\Delta(\mathcal{F}))/(2r)$, and collecting terms gives the identity. □

For $\mathcal{F}=\phi^*E$ one has $\Delta(\mathcal{F})=0$ and $c_1(\mathcal{F})\equiv(\deg E)F$, and the proposition reduces to Theorem 4.1. When $\Delta(K)\ge 0$, the only term that can make the left-hand side negative is $\Delta(\mathcal{F})>0$, which requires a bundle that is not pulled back from the base. In the arithmetic setting, this points back to Parshin's program of an arithmetic Bogomolov–Miyaoka–Yau inequality [11, 9] and to arithmetic inequalities of Bogomolov–Gieseker type [10]. Soulé [12] followed Miyaoka's approach through rank-two extensions; he notes that the arithmetic second Chern number "contains an archimedean summand – an integral over the set of complex points of X – which is not easy to bound from above", and he obtains lower bounds for some successive minima rather than an upper bound for $\overline{\omega}^2$. We draw no further conclusion from this.

8 Concluding Remarks

Within the class of arguments considered here, all available positivity is of one kind: the intersection product of a nef divisor with an effective divisor is nonnegative. In every case treated above, this positivity enters the leading coefficient with the sign that makes the margin nonpositive, whereas a PMB argument requires a positive margin. For pencils, this is established by Theorem 3.1 in rank two and by Corollary 5.2 for semistable ϕ_*L in any rank. What remains open in this direction is the arithmetic anomaly question for unstable ϕ_*L of rank $r\ge 3$ (Open problem 5.5). For [13], version 1, the consequence is limited and concrete: Proposition 9 vi) is false, inequality (36) has a negative leading coefficient, and the proof of Theorem 1 given there is incomplete.

AI Disclosure

AI systems were used extensively in all phases of this work: conception and research discussion, literature and source work, mathematical analysis and proof development, creation and

checking of the computational scripts, execution and evaluation of the computational runs, text, translation, and typesetting.

Quality assurance: use of several independent models from different providers with counter-checks and adversarial AI reviews; traceable, versioned computational scripts with stored result files and checksums; continuously maintained proof and register documents; strict claim hygiene (assertions only with documented evidence, addenda instead of silent corrections); changelogs across all versions.

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